

Classification of Complex Algebraic Varieties

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Outline of the talk

1 Introduction

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- 1 Introduction
- 2 Irreducible subsets

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Polynomial equations

Algebraic Geometry is concerned with the study of solutions of polynomial equations; say n -equations in m -variables

$$\begin{cases} P_1(x_1, \dots, x_m) = 0 \\ \dots \\ P_n(x_1, \dots, x_m) = 0. \end{cases}$$

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$$P(x) = a_d x^d + a_{d-1} x^{d-1} + \dots + a_1 x + a_0 \quad a_d \neq 0$$

(a polynomial of degree d in x).

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- If $a_i \in \mathbb{C}$, then by the Fundamental Theorem of Algebra, there exist $c_i \in \mathbb{C}$ such that $P(x)$ factors as

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- Therefore a polynomial of degree d always has exactly d solutions or roots (when counted with multiplicity).
- If one is interested in solutions that belong to $\mathbb{R}$ (or $\mathbb{Q}$, or $\mathbb{Z}$ etc.), then the problem is much more complicated, but at least we know that there are at most d solutions.

Complex solutions

Throughout this talk I will always look for complex solutions $(z_1, \dots, z_m) \in \mathbb{C}^m$ to polynomial equations

$$P_1(x_1, \dots, x_m) = 0 \quad \dots \quad P_n(x_1, \dots, x_m) = 0$$

where $P_i \in \mathbb{C}[x_1, \dots, x_m]$.

The focus is on understanding the qualitative behaviour of these solutions.

Projective space

- It is more convenient to work on a compactification $\mathbb{C}^N \subset \mathbb{P}_{\mathbb{C}}^N$ where $\mathbb{P}_{\mathbb{C}}^N = \{\mathbb{C}^{N+1} - (0, \dots, 0)\} / \{\mathbb{C}^*\}$ is projective N -space.

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- We have that $\mathbb{P}_{\mathbb{C}}^0$ is a point, $\mathbb{P}_{\mathbb{C}}^1 \cong S^2$ is the Riemann sphere and $\mathbb{P}_{\mathbb{C}}^N = \mathbb{C}^N \cup \mathbb{P}_{\mathbb{C}}^{N-1}$.

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- We then replace $P(x_1, \dots, x_N)$ by the corresponding homogeneous polynomial in $\mathbb{C}[x_0, \dots, x_N]$. Eg.
 $x_1^2 + x_2^3 \Rightarrow x_0 x_1^2 + x_2^3$.

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- We then have:

Bezout's Theorem

Theorem (Bezout's Theorem)

Let $P_1(x_0, \dots, x_n), \dots, P_n(x_0, \dots, x_n)$ be n homogeneous polynomials of degrees $d_1, \dots, d_n$. If the set

$$\bigcap_{i=1}^n \{P_i(x_0, \dots, x_n) = 0\} \subset \mathbb{P}_{\mathbb{C}}^n$$

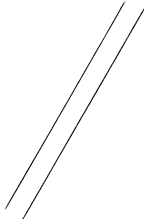
is finite, then it consist of exactly $d_1 \cdots d_n$ points (counted with multiplicity).

Lines in a plane

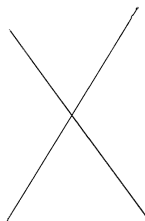
CASE 1
 $P_1(x, y) = c P_2(x, y)$



CASE 2
 $P_1(x, y) = c P_2(x, y) + b$

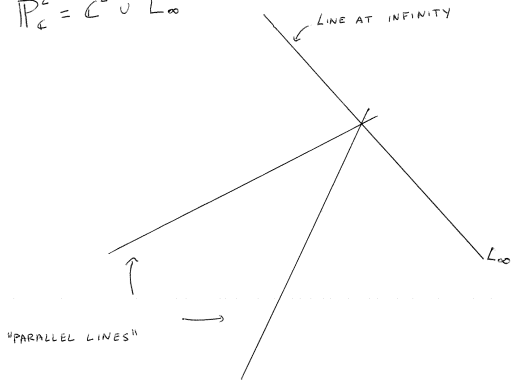


CASE 3

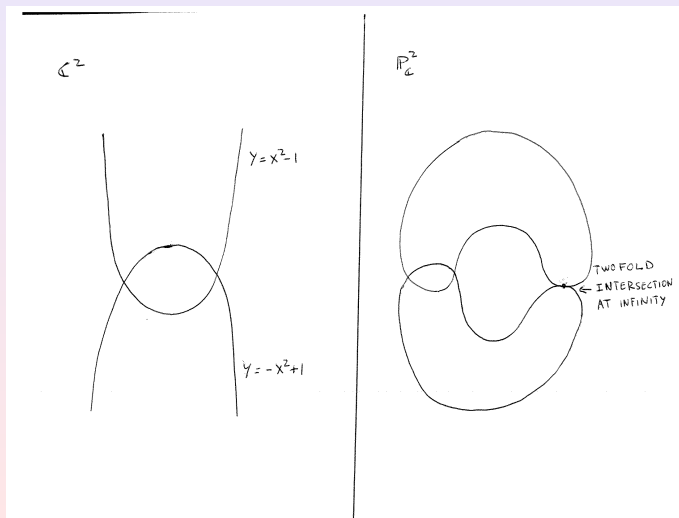


Lines in a projective plane

$$\mathbb{P}^2 = \mathbb{C}^2 \cup L_\infty$$



Quadrics in a plane



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Irreducible subsets

- In general a **Zariski closed subset** of $\mathbb{P}_{\mathbb{C}}^m$ is a set of the form

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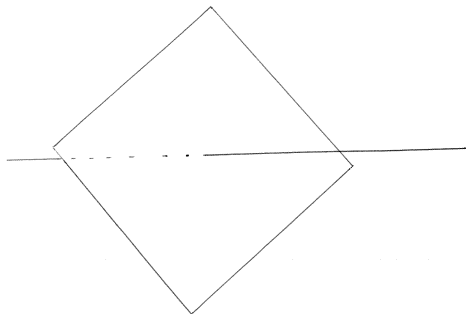
- A Zariski closed subset of $\mathbb{P}_{\mathbb{C}}^m$ is **irreducible** if it can not be written as the union of 2 proper Zariski closed subsets.
- Any Zariski closed subset is a finite union of irreducible Zariski closed subsets.
- If $P(x, y, z) \in \mathbb{C}[x, y, z]$ is homogeneous of degree $r > 0$, then

$$X = \{P(x, y, z) = 0\}$$

is irreducible if and only if $P(x, y, z)$ does not factor.

Reducible set

REDUCIBLE ZARISKI CLOSED SUBSET OF $\mathbb{P}^3$



Irreducible subsets II

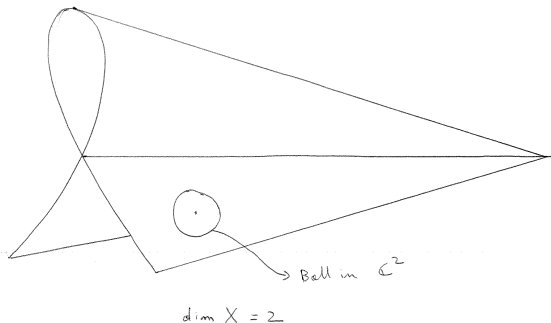
- If X is an infinite irreducible Zariski closed subset, then there is an integer $d > 0$ such that for most points $x \in X$, one can find an open neighborhood $x \in U_x \subset X$ where U_x is analytically isomorphic to a ball in $\mathbb{C}^d$.

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- We then say that x is a **smooth** point and that X has **dimension** d .

Irreducible set of dimension 1

SINGULAR SET IN $\mathbb{P}^3_{\mathbb{C}}$



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- For “most” systems of homogeneous polynomial equations $P_1, \dots, P_r \in \mathbb{C}[X_0, \dots, X_N]$, the set $X = \text{Zeroes}(P_1, \dots, P_r) \subset \mathbb{P}_{\mathbb{C}}^N$ is smooth.

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- For “most” systems of homogeneous polynomial equations $P_1, \dots, P_r \in \mathbb{C}[X_0, \dots, X_N]$, the set $X = \text{Zeroes}(P_1, \dots, P_r) \subset \mathbb{P}_{\mathbb{C}}^N$ is smooth.
- By a deep result of Hironaka, there exists a **desingularization** of X , that is a morphism $f : Y \rightarrow X$, such that Y is everywhere smooth and f is the identity over any smooth point of X .

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- There is only one such curve $X = \mathbb{P}_{\mathbb{C}}^1$.
- When $g = 1$ we have a 1-parameter family of **elliptic curves** $X_t = \{y^2 - x(x-1)(x-t) = 0\}$.

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- We would like to find a natural (canonical) description for $X \subset \mathbb{P}_{\mathbb{C}}^m$.

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- If L is a line bundle on X and $s_0, \dots, s_m$ is a basis of $H^0(X, L)$, then there is a map $\phi_L : X \dashrightarrow \mathbb{P}_{\mathbb{C}}^m$ given by

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- To define an embedding of X in $\mathbb{P}_{\mathbb{C}}^m$, we must find a line bundle L whose sections distinguish different points and tangent directions.

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Theorem

Let X be a curve of genus $g \geq 2$, then the sections of $H^0(X, \omega_X^{\otimes 3})$ define an embedding of degree $6g - 6$

$$X \hookrightarrow \mathbb{P}_{\mathbb{C}}^{5g-6}.$$

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- One can make this more precise by using Hilbert schemes.
- Ultimately one can construct a moduli space (irreducible of dimension $3g - 3$) and its compactification.
- This moduli space “classifies” all curves of genus g .

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- In coordinates this may be described as follows. If $r_0, \dots, r_m$ are generators of $R(\omega_X)$ and $K = \ker(\mathbb{C}[x_0, \dots, x_m] \rightarrow R(\omega_X))$, then $X \subset \mathbb{P}_{\mathbb{C}}^m$ is defined by the equations in K .

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- The **Kodaira dimension** of X is $\kappa(X) = \text{tr.deg.}_{\mathbb{C}} R(\omega_X) - 1$.
- We have $\kappa(X) = \max_{k \geq 0} \{\dim \phi_{\omega_X^{\otimes k}}(X)\}$ so that $\kappa(X) \in \{-1, 0, 1, \dots, d\}$.

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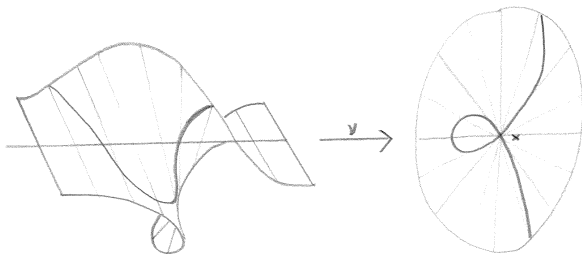
- The morphism ν may be viewed as a surgery that replaces the point x by a rational curve $E \cong \mathbb{P}^1 = \mathbb{P}(T_x X)$.
- The exceptional curve E is called a -1 **curve** since

$$\omega_{\tilde{X}} \cdot E = \deg(\omega_{\tilde{X}}|_E) = -1.$$

Blow up

$$\mathrm{Bl}_x(X) \subset \mathbb{A}^2 \times \mathbb{P}^1_{\mathbb{A}}$$

$$X = \mathbb{A}^2$$



$$\{(x_1, x_2)[y_1 : y_2] \mid x_1 y_2 = x_2 y_1\}$$

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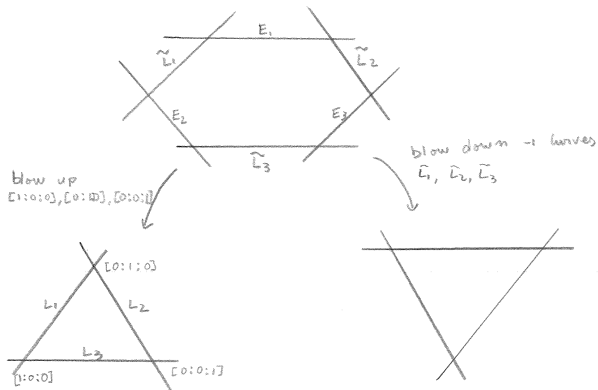
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- Remark: The canonical ring $R(\omega_X)$ is a birational invariant.

Birational map of the plane

$$[x:y:z] \dashrightarrow [1/x : 1/y : 1/z] = [yz : xz : xy]$$



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- A **minimal surface** is a surface that contains no -1 curves.
- By Castelnuovo's Theorem, given any surface X there exists a birational morphism to a minimal surface $X \rightarrow X_{\min}$ (given by finitely many blow downs).

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- If $\kappa(X) \geq 0$, then the minimal model $X \rightarrow X_{min}$ is unique and $\omega_{X_{min}}$ is **nef**. This means that for any curve $C \subset X_{min}$ we have

$$\omega_{X_{min}} \cdot C \geq 0.$$

Birational geometry of surfaces V

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- Conclusion: After performing a finite sequence of geometrically meaningful operations (contracting -1 curves) we obtain a birational model X_{min} on which $\omega_{X_{min}}$ is positive (nef and semiample).

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- If $\kappa(X) = 1$, then there is a morphism $X_{min} \rightarrow B$ whose general fiber F is a curve of genus 1; $\omega_{X_{min}} \cdot F = 0$.
- It is impossible to classify all surfaces with $\kappa(X) = 2$. However we have a good qualitative understanding of their geometry.

Birational geometry of surfaces VII

- If $\kappa(X) = 2$, then $\phi_{\omega_{X_{min}}^{\otimes 5}} : X_{min} \rightarrow \mathbb{P}^N$ is a morphism whose image $X_{can} \cong \text{Proj } \bigoplus_{m \geq 0} H^0(\omega_X^{\otimes m})$ is obtained from X_{min} by contracting rational curves $E \cong \mathbb{P}^1$ with $\omega_{X_{min}} \cdot C = 0$.

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- The construction does not work for K3's (there are infinitely many components of the moduli space).

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- Is there a most natural desingularization?
- Answer: the **minimal desingularization**.
- I.e. a map $X' \rightarrow X$ such that X' is smooth and any desingularization $Y \rightarrow X$ factors as $Y \rightarrow X' \rightarrow X$.

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- So, from the moduli space for Y_{can} we obtain a moduli space for Y_{min} .

Outline of the talk

- 1 Introduction
- 2 Irreducible subsets
- 3 Curves
- 4 Surfaces
- 5 Higher dimensions**

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- This is implied by the finite generation of the canonical ring $R(\omega_X)$.
- Recently there has been substantial progress in higher dimensions.

Higher dimensions

Theorem (Birkar-Cascini-Hacon-M^cKernan, Siu)

The canonical ring

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It follows that canonical models $X_{can} = \text{Proj} R(\omega_X)$ exist.
We can also construct minimal models (in most cases).

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In the next lecture I will discuss in some detail the MMP. In particular I will introduce the new elementary birational operations required to construct minimal models (flips and flops).